

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each.

Hardware Issue	Typical Symptom
No display on monitor	Blank screen, monitor shows "No Signal," and the computer may produce no POST (Power-On Self-Test) beep.
Faulty RAM (Memory)	Computer fails to boot, continuous or repeated beep codes, random crashes, or frequent Blue Screen of Death (BSOD).
Hard drive/SSD failure	Operating system does not load, "Boot device not found" error, slow performance, clicking noises (HDD), or missing files.
Power Supply Unit (PSU) failure	PC does not power on at all, no lights or fans, or the system suddenly shuts down or restarts unexpectedly.
Overheating CPU or cooling fan failure	Computer becomes very hot, fan runs loudly or stops spinning, frequent freezing, automatic shutdowns, or poor performance.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

1. Verify Power Source and Cables

- Even if your laptop's indicator light comes on, it may be receiving insufficient power.
- **Wall Outlet:** Plug a different working device (like a lamp or phone charger) into the outlet to verify it supplies electricity.
- **Power Strips:** If using a surge protector, bypass it and plug the laptop's charger directly into the wall to rule out strip failures.
- **Connection Points:** Check all connection points along your charging cord. Ensure the AC adapter (power brick) is securely attached to the wall-side cable and firmly seated into the laptop

- **Visual Inspection:** Run your fingers along the entire length of the charging wire. Feel for any kinks, deep bends, or warped areas, and inspect the ends for frayed wiring or bent connectors.

2. Inspect the Charging Port and Battery

- Physical blockages and degraded components are common culprits.
- **Clear Debris:** Look directly into the laptop's charging port (jack) using a flashlight. Dust or lint can prevent the charging pin from making solid contact. If debris is present, turn off the laptop and gently clean the contacts using a wooden or plastic toothpick.
- **Check for Damage:** Gently plug in the charging cable. It should sit snugly. If the connector feels loose, wobbly, or falls out easily, the internal charging port or DC jack may be physically broken.
- **Removable Batteries:** If your laptop features a removable battery, take it out and inspect it for signs of cracking or swelling. Re-seat it firmly, ensuring the locking latches engage properly.

3. Perform a Hard Power Reset

- Lingering static electricity or a frozen Embedded Controller (EC) can prevent a laptop from recognizing the power adapter.
- **Remove Power:** Shut down the computer completely and unplug the AC adapter. Remove the battery if it is externally removable.
- **Discharge:** Press and hold down the power button for 25 to 30 seconds.
- **Restart:** If applicable, put the battery back in. Plug the AC adapter into the laptop, and turn the PC on to see if it registers the charge.

4. Hardware Limit Check and Alternative Charging

- Sometimes the issue is an active charge-threshold setting or an incompatible charger.
- **Charge Thresholds:** Many manufacturers include battery-saving software that limits charging (e.g., stopping at 80% to preserve battery lifespan)
- **USB-C Functionality:** If you are using a USB-C charger, verify it is plugged into a charging-supported port. Many laptops feature multiple USB-C ports, but only designated ones (often marked with a lightning or power plug icon) are wired for Power Delivery.
- **Try a Different Charger:** Test a compatible, known-working charger (matched to your laptop's required wattage) to determine if your original power brick is faulty.

3. **Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB)**

Virtual USB Charger Test

1. Setup and Procedure:

- **Tool:** Digital Multimeter (DMM).
- **Setting:** The dial is set to measure direct current (DC) voltage, specifically the 20V range (often denoted by a 'V' with a solid and dashed line above it).
- **Probing:** The red (positive) probe is connected to Pin 1 (VCC / +5V), and the black (negative) probe is connected to Pin 4 (Ground) on the outer edges of the standard USB-A connector.

2. Simulated Measurement:

- **Measured Voltage: 5.04V DC**

3. Analysis and Conclusion:

- **Expected Range:** 4.75V to 5.25V.
- **Conclusion:** The simulated measurement of **5.04V** is safely within the expected range.
- A standard USB charger aims for a flat 5.00V output. The acceptable 4.75V to 5.25V window exists to account for minor manufacturing tolerances and the natural voltage drop that happens as current travels down a USB cable.
- *If it were too low (under 4.75V):* The device might charge incredibly slowly or fail to recognize the charger entirely.
- *If it were too high (over 5.25V):* You run the risk of frying the charging circuitry inside your phone.

4. **A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?**

Setup and Testing Procedure

To test a disconnected ATX power supply, you have to trick it into turning on without a motherboard.

1. Prep the PSU:

- Turn off the switch on the back of the PSU and unplug it from the wall.
- Disconnect all PSU cables from your motherboard, GPU, and drives.
- Plug the PSU back into the wall, but leave the back switch off for now.

2. Jump the Power Supply (The "Paperclip Test"):

- Locate the 24-pin ATX connector.
- Use a bent paperclip or a dedicated PSU jumper bridge to connect the **PS_ON** pin (usually a green wire, Pin 16) to any **Ground/COM** pin (black wire, like Pin 15 or 17).
- Flip the switch on the back of the PSU to the "On" position. The PSU fan should spin up.

3. Configure the Multimeter:

- Set the Digital Multimeter (DMM) dial to measure **DC Voltage**, specifically the **20V range** (often marked with a 'V' and a solid/dashed line).

4. Probe the Connector:

- Insert the **black (negative) probe** into any socket with a black Ground wire.
- Insert the **red (positive) probe** into a socket with a **yellow wire** (the +12V rail).

II. Simulated Measurement

- **Target Rail:** +12V DC
- **Simulated Measured Voltage:** 11.24V DC

III. Diagnostic Breakdown: What a Low Reading Means

- The ATX specification dictates that the +12V rail must remain within a $\pm 5\%$ tolerance. This means a healthy 12V rail should always measure between **11.40V and 12.60V**.
- If your reading is below 11.5V (like our simulated **11.24V**), it indicates a **failing power supply**. Specifically:

- **Internal Component Degradation:** The voltage regulators or capacitors inside the power supply are likely degrading and failing to hold a stable charge.
- **The "No Load" Red Flag:** Because you are testing the ATX connector disconnected from the PC, the PSU is operating under *zero load*. If it cannot maintain a baseline 12V when doing absolutely no work, the voltage will plummet even further the second your CPU or GPU demands power.
- **Link to Random Shutdowns:** When you launch a game or open a heavy application, the power demand spikes. Because the 12V rail is already heavily under-performing, the sudden spike causes the voltage to dip so low that the motherboard's Under-Voltage Protection (UVP) kicks in to save your components, instantly shutting down the PC.
- **The Verdict:** If you pull a reading below 11.5V on a disconnected ATX connector, the PSU is dead or dying and must be replaced immediately.